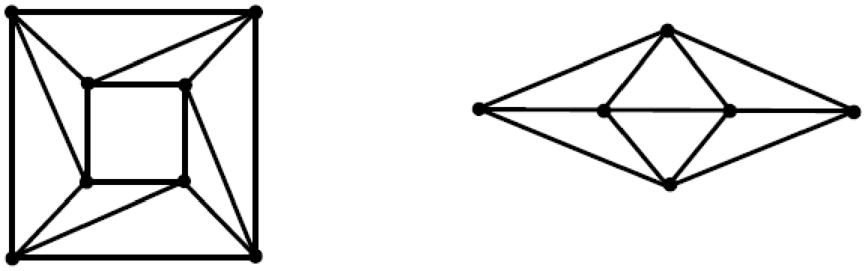
FORMAL GRAPH THEORY: COLOURING **13/03/23**

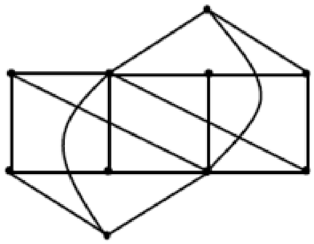
Some key definitions you need to know:

- A **vertex colouring** of a (simple) graph is an assignment of colours to the vertices of a graph so that no two adjacent vertices are the same colour. “Colours” can literally be colours, or simple labels {1,2,3,…} or {a,b,c,…}.
- A graph is said to be ***k*-coloured** if it can be (vertex) coloured with *k* colours.
- The **vertex chromatic number** of a graph is the minimum value of *k* for which the graph can be *k*-coloured.

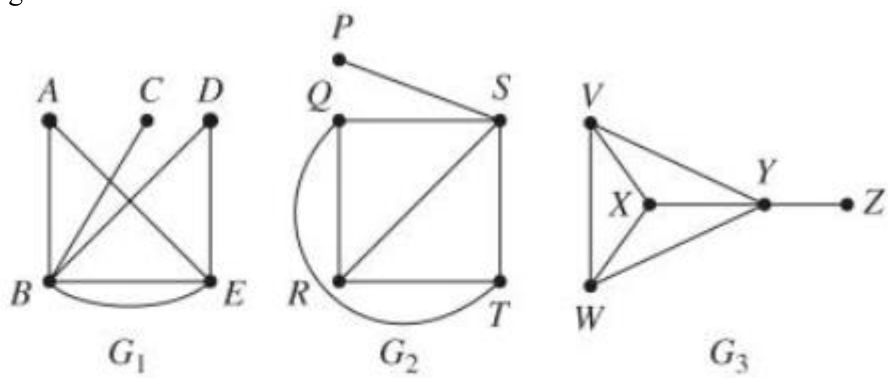
1. What are the vertex chromatic numbers of these two graphs?



2. What is the vertex chromatic number of a complete graph of order n?
3. What is the vertex chromatic number of a tree?
4. What is the vertex chromatic number of a bipartite graph?
5. What can you say about a graph with vertex chromatic number 2?
6. Is this graph bipartite?



7. Use a colouring argument to identify the two graphs from these three which are isomorphic. List corresponding vertices.



8. Prove that a graph is bipartite if and only if every cycle in the graph has even length. Why is every tree bipartite?

9. Prove that a graph with chromatic number 3 must have an odd cycle.

10. Does there exist a bipartite graph with degree sequence 5, 5, 5, 4, 4, 3, 3, 3, 1, 1, 1, 1?

11.
   a. Show that a graph with 17 vertices and 73 edges cannot be bipartite
   b. Show that a bipartite graph of order n can have at most $\left\lfloor \frac{n^2}{4} \right\rfloor$ edges.

12.
   a. How many ways are there to sit three clowns and three politicians at a round table, if the clowns and politicians must alternate?
   b. How many different cycles of length 6 are there in the bipartite graph $K_{3,3}$?

13. A company is shipping chemicals A to H, some of which react dangerously together and so cannot be shipped in the same container, as shown below:

chemical	cannot be shipped with
A	B, C, D, E
B	A, C, D, E
C	A, B, F, G, H
D	A, B, G, H
E	A, B
F	C, G, H
G	C, D, F, H
H	C, D, F, G

   a. Draw a graph to represent this problem, with chemicals as vertices, with edges joining every pair of chemicals which cannot be shipped together.

   b. The company wishes to minimise the number of separate shipping containers used. Explain how to interpret this in terms of vertex colouring and hence decide, with justification, what the minimum number of containers is.

14. School exams are being timetabled. Six students will take papers as follows.

student 1: Ma, Bi, Ch;
student 2: Ma, Ch, La;
student 3: En, Hi, Gy;
student 4: Ma, Bi, Fr;
student 5: La, En, Fr;
student 6: Bi, Ch, Gy.

Model this information using a graph, and solve a vertex colouring problem to find the minimum number of exam sessions needed to avoid clashes.